

Derivation of the Minimal Length Scale L_0

from the T0 Lagrangian without Postulates

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<https://github.com/jpascher/T0-Time-Mass-Duality>

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Abstract

The T0 theory postulates neither the parameter ξ nor the minimal length scale $L_0 = \xi \cdot \ell_p$. Both emerge as necessary consequences of the self-consistency of the T0 Lagrangian. This document presents the complete derivation chain in five steps: (1) Lagrangian structure with time-mass duality, (2) self-consistency condition and 10^{-4} scaling, (3) uniqueness of ξ from the e-p- μ system, (4) gravitational constant G from ξ , (5) minimal length scale $L_0 = \xi \cdot \ell_p$.

.1 Step 1: The T0 Lagrangian

Standard Model Starting Point

The Standard Model Lagrangian for a lepton ψ_ℓ reads:

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell \quad (1)$$

with $D_\mu = \partial_\mu + ieA_\mu$ and constant m_ℓ .

Time-Mass Duality as Fundamental Principle

In T0 theory the fundamental relation holds:

$$\tilde{T}(x, t) \cdot m(x, t) = 1 \quad (2)$$

This is not an additional postulate but a structural condition: every mass carries an intrinsic time field. This necessarily implies a dynamic mass field:

$$m(x, t) = m_0 + \Delta m(x, t) \quad (3)$$

Complete T0 Lagrangian

Complete T0 Lagrangian

$$\mathcal{L}_{\text{T0}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell + \frac{1}{2}(\partial_\mu \Delta m)(\partial^\mu \Delta m) - \frac{1}{2}m_T^2 \Delta m^2 + \xi m_\ell \bar{\psi}_\ell \psi_\ell \Delta m \quad (4)$$

The Three Key Formulae

Coupling strength (mass-proportional):

$$g_T^\ell = \xi \cdot m_\ell \quad (5)$$

Time field mass:

$$m_T = \frac{\lambda}{\xi} \quad (6)$$

Vertex factor:

$$\bar{\psi}_\ell \psi_\ell \Delta m \longrightarrow -i \xi m_\ell \quad (7)$$

Time field propagator:

$$\Delta m(k) \longrightarrow \frac{i}{k^2 - m_T^2 + i\epsilon} \quad (8)$$

.2 Step 2: Self-Consistency Condition

Why the Lagrangian Forces ξ

The coupling term $g_T^\ell = \xi \cdot m_\ell$ is not freely choosable. The self-consistency of the Lagrangian requires:

- The theory remains free of divergences at high energies.
- The known particle masses are reproduced.
- The mass hierarchy between generations is geometrically describable.

The 10^{-4} Factor from QFT Loop Structure

The loop suppression in four spacetime dimensions yields:

$$\frac{1}{16\pi^3} \approx 2.01 \times 10^{-3} \quad (9)$$

Combined with the Higgs parameters from the T0 Lagrangian:

$$(\lambda_h)^2 \cdot \frac{v^2}{m_h^2} = (0.129)^2 \times \frac{(246.2)^2}{(125.1)^2} \approx 0.0647 \quad (10)$$

10^{-4} from Spacetime Dimensionality

$$\xi_4 \sim (10^{-1})^4 = 10^{-4} \quad (11)$$

This factor is not postulated — it follows from the dimensionality of spacetime.

The 4/3 Factor from Tetrahedral Geometry

The factor 4/3 emerges from the harmonic structure of three-dimensional space:

$$\frac{4 \text{ vertices}}{3 \text{ edges per vertex}} = \frac{4}{3} \quad (\text{perfect fourth}) \quad (12)$$

The tetrahedron contains both fundamental harmonic intervals:

$$\frac{3}{2} \times \frac{4}{3} = 2 \quad (\text{octave}) \quad (13)$$

.3 Step 3: Uniqueness of ξ

The e-p- μ System as Uniqueness Proof

From the three fundamental mass ratios:

$$R_{pe} = \frac{m_p}{m_e} = 1836.15267343 \quad (14)$$

$$R_{\mu e} = \frac{m_\mu}{m_e} = 206.7682830 \quad (15)$$

$$R_{p\mu} = \frac{m_p}{m_\mu} = 8.880 \quad (16)$$

With the mass scaling $m_p/m_e = 245 \times (4/3)^\kappa$ it follows:

$$\kappa = \frac{\ln(R_{pe}/245)}{\ln(4/3)} = \frac{\ln(1836.15/245)}{\ln(1.3333)} \approx 7.000 \quad (17)$$

$\kappa = 7$ is the Only Consistent Solution

Exponent κ	R_{pe} prediction	Consistent	Error
$\kappa = 6$	$245 \times (4/3)^6 = 1376.6$	×	25.0%
$\kappa = 7$	$245 \times (4/3)^7 = 1835.4$	✓	0.04%
$\kappa = 8$	$245 \times (4/3)^8 = 2447.2$	×	33.3%

Table 1: $\kappa = 7$ is the only consistent solution

ξ as Emergent Constant

$$\xi = \frac{4}{30000} = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4} \quad (18)$$

ξ is not a postulate — it is the only self-consistent solution of the e-p- μ system.

.4 Step 4: Gravitational Constant G from ξ

Fundamental T0 Gravitational Relation

The T0 Lagrangian implies the fundamental geometric relation:

$$\xi = 2\sqrt{G \cdot m_{\text{char}}} \Rightarrow G = \frac{\xi^2}{4 m_{\text{char}}} \quad (\text{T0 natural units}) \quad (19)$$

The complete SI value requires explicit dimensional and unit conversion factors (Document 012):

$$G_{\text{SI}} = \frac{\xi^2}{4 m_e} \times C_{\text{dim}} \times C_{\text{conv}} \times K_{\text{frak}} \quad (20)$$

where C_{dim} is the dimensional correction for the transition from T0 natural to physical units, C_{conv} the SI conversion factor, and K_{frak} the fractal correction. Numerically:

$$G_{\text{SI}} \approx 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2 \quad (21)$$

Deviation from CODATA-2018: $< 0.01\%$. The complete derivation of all factors is in Document 012.

.5 Step 5: Minimal Length Scale L_0

Planck Length from G and ξ

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} = 1.616 \times 10^{-35} \text{ m} \quad (22)$$

Since G is derived from ξ , ℓ_P follows completely from ξ .

L_0 as Sub-Planck Scale

The time-energy duality $r_0(E) \leftrightarrow E$ in natural units requires a minimal length scale. The coupling term $g_T^\ell = \xi \cdot m_\ell$ scales with ξ — below L_0 the concept of “distance” loses its physical meaning, because the time field fluctuations Δm dissolve the spacetime structure itself.

Minimal Length Scale

$$L_0 = \xi \cdot \ell_P = \frac{4}{30000} \times 1.616 \times 10^{-35} \text{ m} = 2.155 \times 10^{-39} \text{ m} \quad (23)$$

Physical Meaning of L_0

L_0 marks the scale where TO geometric effects become dominant. Below L_0 :

- All vacuum fluctuations are fully effective (Casimir effect).
- The perturbative expansion loses its validity.
- The fractal correction K_{frak} becomes structurally necessary.

.6 Summary: Complete Derivation Chain

Derivation Chain without Postulates

$$\begin{aligned}
 &\text{Lagrangian structure } (\tilde{T} \cdot m = 1) \\
 &\Downarrow \\
 &\text{Self-consistency requires } g_T = \xi \cdot m \\
 &\Downarrow \\
 &10^{-4} \text{ factor from 4D spacetime, } 4/3 \text{ factor from tetrahedral geometry} \\
 &\Downarrow \\
 &\kappa = 7 \text{ only solution of the e-p-}\mu \text{ system} \\
 &\Downarrow \\
 &\boxed{\xi = 4/30000} \quad (\text{not a postulate — emergent constant}) \\
 &\Downarrow \\
 &G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \times K_{\text{frak}} \quad \Rightarrow \quad \ell_P = \sqrt{\hbar G / c^3} \\
 &\Downarrow \\
 &\boxed{L_0 = \xi \cdot \ell_P = 2.155 \times 10^{-39} \text{ m}} \quad (24)
 \end{aligned}$$

.7 Connection to the SI System

The Unavoidable Circularity

All results in this document — and in the companion documents 181 and 182 — are initially **ratio-based**: ξ is dimensionless, $L_0/\ell_P = \xi$ is a pure ratio, $G = \xi^2/(4m_e)$ holds in natural units.

As soon as one wishes to state absolute SI values — i.e. $L_0 = 2.155 \times 10^{-39}$ m rather than just $L_0 = \xi \cdot \ell_P$ — at least **one** external connection to the SI system is required.

Apparent Circularity is Unavoidable

It is not possible to derive absolute SI values from a purely dimensionless parameter like ξ without establishing somewhere a connection to the SI unit definition. This connection is not a weakness of the theory — it is an epistemological necessity of every physical theory.

Possible Entry Points into the SI System

The connection can be established via various quantities, all of which are equivalent:

- **Via α :** The fine-structure constant $\alpha = \xi \cdot E_0^2 / (1 \text{ MeV})^2$ is dimensionless and experimentally very well known. It provides via $E_0 = \sqrt{m_e \cdot m_\mu}$ the connection to the lepton masses in MeV.
- **Via m_{char} or m_e :** The electron mass $m_e = 0.511 \text{ MeV}$ is experimentally defined and links ξ to SI units via $G = \xi^2 / (4m_e) \times C_{\text{dim}} \times C_{\text{conv}} \times K_{\text{frak}}$.
- **Via ℓ_P :** The Planck length $\ell_P = \sqrt{\hbar G / c^3}$ is experimentally accessible and gives directly $L_0 = \xi \cdot \ell_P$ in metres.

Which Entry is Fundamental?

In T0/FFGFT the most natural entry is via α , because:

- α is itself derived from ξ : $\alpha = \xi \cdot E_0^2 / (1 \text{ MeV})^2$
- α is dimensionless — the transition to SI units occurs only through the choice of unit scale (MeV)
- The 2019 SI reform implemented exactly this calibration, which is consistent with the T0 geometry (Document 013)

Epistemological Classification

The derivation chain in this document is internally consistent and free of postulates. The only external input is the choice of an SI unit scale — for example via m_e in MeV or ℓ_P in metres. This input is not a circularity of the theory, but the necessary calibration of every physical theory to the world of measurement. Treated in detail in Document 056 (Universal Derivation) and Document 101 (Circularity of Constants).